

SATYAJIT SAHOO

Full Stack Developer | AI/ML Engineer | DevOps Engineer

+91-7787998637 | satyajitofficial4@gmail.com | [linkedin.com/in/satyajitsjs](https://www.linkedin.com/in/satyajitsjs) | github.com/satyajitsjs

PROFESSIONAL PROFILE

Results-driven **Full Stack Developer** and **AI/ML Engineer** with **3+ years** of professional experience designing, developing, and deploying scalable web applications and intelligent systems across enterprise-grade products. Deep expertise in **Python, Django REST Framework, React.js, Next.js, TypeScript**, and **cloud-native DevOps** on Azure and AWS. Proven track record delivering production-ready solutions across **IoT, e-commerce, supply chain, machine learning, computer vision**, and **EdTech**. Experienced in leading cross-functional teams, CI/CD automation, and agile delivery. Strong communicator in English; eager to contribute to innovative, high-impact engineering teams internationally.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, Java, C, C#, PHP, Go, Ruby
Frameworks	Django, Django REST Framework, React.js, Next.js, Laravel, Ext JS, Celery
Frontend & UI	HTML5, CSS3, Tailwind CSS, Bootstrap, jQuery, Ajax, Figma, Responsive Web Design
AI / ML / CV	Machine Learning, Deep Learning, Computer Vision, CNN, LSTM, Time Series Forecasting, Transfer Learning (ResNet, EfficientNet), TensorFlow, Keras, Scikit-learn, OpenCV
Data & Analytics	NumPy, Pandas, Power BI, Feature Engineering, Technical Indicators (RSI, MACD)
Databases	MySQL, MongoDB, PostgreSQL, SQLite3, Redis, Firebase Firestore
DevOps & Cloud	AWS, Azure, Azure DevOps, Docker, Docker Compose, GitHub Actions, CI/CD Pipelines, Ansible, Terraform, Cloudflare
Auth & Security	Firebase Auth, Keycloak, Google OAuth, JWT, Zod Validation, RBAC
IoT & Protocols	EMQX, MQTT, LoRaWAN, Sensor Integration, Real-time Data Streaming
Methodology	Agile, Scrum, REST APIs, Microservices, OOP, Data Structures & Algorithms

WORK EXPERIENCE

Assistant Manager – Software Engineering <i>Landmark Group</i>	Nov 2025 – Present <i>Full Stack Supply Chain Enterprise Software</i>
- Leading full-stack development of FabricFlow, an enterprise fabric supply chain platform covering end-to-end procurement, fabric customisation, inventory management, and multi-tier retailer distribution. - Architected a scalable Next.js + TypeScript frontend with Zod schema validation on all customisation and procurement forms, eliminating runtime data errors and ensuring strict type safety at scale. - Designed MongoDB data models and aggregation pipelines for procurement analytics, inventory tracking, customisation workflow states, and real-time retailer management dashboards. - Integrated Firebase for real-time authentication, push notifications, and live order status updates; enforced RBAC with JWT-based access control across vendor, admin, and retailer portals. - Built and documented RESTful APIs for vendor onboarding, procurement lifecycle, customisation approvals, and dispatch coordination – improving cross-team integration and reducing manual handoffs by 50%. - Translated complex supply chain requirements into scalable technical architecture; led code reviews and maintained high engineering quality standards across the development team.	
Software Engineer <i>Smartx Technology</i>	Nov 2024 – Nov 2025 <i>Backend DevOps EdTech ML Integration</i>
- Designed, managed, and optimised Azure DevOps CI/CD pipelines and infrastructure automation workflows, reducing deployment times and improving release reliability across all active projects. - Collaborated with data scientists to containerise and deploy machine learning models to production using Docker and Azure, ensuring seamless ML-to-product integration. - Led and mentored 6+ concurrent engineering projects across cloud infrastructure, DevOps automation, and backend systems – delivering on-time with high technical quality. - Architected AptPath LMS, a multi-role learning management platform with Django/Python REST APIs for course management, student progress tracking, and real-time academic analytics. - Built an internal Task Management System with Django Admin, Celery task queues, Firebase Push Notifications, Google OAuth, and Greythr API for seamless HR data synchronisation. - Championed engineering best practices – modular API design, automated testing pipelines, and structured code reviews – raising overall team productivity and code quality.	
Software Associate <i>Tritech Software Private Limited</i>	Aug 2022 – Oct 2024 <i>Full Stack E-Commerce Real-time Analytics</i>
- Engineered a production-grade e-commerce platform with Python, Django, JavaScript, jQuery, and Ajax, delivering an 85% uplift in online marketing conversions within six months of launch. - Built a real-time analytics dashboard with React.js and Django REST API, improving data processing efficiency by 60% and enabling live business intelligence reporting for business stakeholders.	

- Optimised **MongoDB** aggregation queries and **Redis** caching strategies, achieving **40% faster** data retrieval under peak concurrent user load.
- Implemented end-to-end **CI/CD pipelines** via Azure DevOps, automating build, test, and deploy workflows – significantly reducing manual release overhead and improving production stability.

KEY PROJECTS

Cryptocurrency Price Prediction System *Python | LSTM | TensorFlow | Django | React.js*

- Built an end-to-end **crypto price forecasting platform** using **LSTM** neural networks trained on historical OHLCV data for BTC, ETH, and altcoins with multi-step future price prediction.
- Engineered feature pipelines with **technical indicators** (RSI, MACD, Bollinger Bands, EMA, SMA) using Pandas and NumPy, significantly boosting model accuracy and signal reliability.
- Developed **Django REST API** with live Binance/CoinGecko market feeds; containerised inference stack with **Docker**; built **React.js dashboard** with candlestick charts and buy/sell signals.
- Implemented continuous model retraining pipeline from live market data, ensuring prediction accuracy improved over time through automated feedback loops and scheduled retraining jobs.

Product Price Direction Predictor – Computer Vision AI *CNN | TensorFlow | OpenCV | Django | Firebase*

- Developed an **AI system** that analyses a product image and classifies whether the price will trend **UP or DOWN** using **CNN deep learning** and transfer learning (ResNet/EfficientNet).
- Fused product image features with historical pricing data for binary classification; improved accuracy through iterative training, validation, and hyperparameter optimisation.
- Exposed model via **Django REST API** for real-time image upload and inference; stored predictions in **MongoDB** for continuous retraining; integrated **Firebase Storage** for image management.
- Built **React.js** frontend for intuitive image upload, real-time prediction results, and confidence score visualisation supporting multiple product categories.

AptPath – Learning Management System (LMS) *Django | Python | Firebase | Keycloak | REST API*

- Designed a scalable **multi-role LMS backend** using Django/Python supporting universities, colleges, teachers, and students on a single high-availability platform.
- Developed comprehensive **REST APIs** for course creation, content delivery, student enrolment, progress tracking, quiz assessments, grading, and result analytics for real-time academic insights.
- Integrated **Firebase** and **Keycloak** for dual-layer authentication with fine-grained **RBAC**, ensuring secure role-appropriate access for administrators, teachers, and students.
- Optimised **database query performance** through strategic indexing and Redis caching, enabling the platform to handle high concurrent loads with consistently low API response times.

LoRa Gateway & Aquaculture IoT Dashboard *Python | Django | React.js | LoRaWAN | MQTT*

- Developed a **LoRaWAN IoT gateway** for long-range, low-power device communication using Python/Django with end-to-end encrypted data transmission and device-level authentication.
- Built a **real-time aquaculture monitoring dashboard** with React.js displaying live sensor feeds for water temperature, pH, and dissolved oxygen for precision fish health management.
- Integrated multi-sensor data pipeline using **MQTT** and **EMQX** message broker for reliable, high-throughput IoT data ingestion from distributed remote field deployments.
- Conducted extensive field testing and RF signal optimisation across large aquaculture sites; implemented automated critical threshold alerting for immediate operational response.

FabricFlow – Enterprise Fabric Supply Chain Platform *Next.js | TypeScript | Zod | MongoDB | Firebase*

- Architected full supply chain system – vendor onboarding, procurement, fabric customisation, inventory, and retailer distribution – with **Next.js + TypeScript** and **Zod**-validated forms.
- Designed **MongoDB** aggregation pipelines for stock analytics and workflow states; integrated **Firebase** for real-time sync and automated order approval and dispatch notifications.

E-Commerce Platform *Django | Python | React.js | Razorpay | Docker*

- Full-featured e-commerce platform with product catalogue, dynamic pricing, **Razorpay** payment gateway (webhooks, refunds), custom admin panel with sales analytics, and Dockerised deployment.

EDUCATION

B.Tech – Computer Science	Bhubaneswar Institute of Industrial Technology	2022	73.48%
Higher Secondary (Science)	Sarala Residential +2 Science College	2018	55.33%

LANGUAGES

English – Professional working proficiency (C1)	Hindi – Native	Odia – Native
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I confirm that all information provided in this CV is accurate and complete to the best of my knowledge.